THE SCHEME THEORETIC IMAGE IS INSENSITIVE TO A SURJECTIVE FLAT COVER OF FINITE PRESENTATION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The scheme theoretic image of a morphism of algebraic stacks, characterised by the closed substacks through which the morphism factors. The author's section on scheme theoretic images, in which the counted statement occurs. Supplies the universal property against which existence and equality are tested.
- The fibre product of a morphism along a surjective flat morphism of finite presentation is again surjective flat of finite presentation, so that it admits an fppf covering by a scheme. Used inside the counted proof (source lines 8768-8776). Produces the covering of the test scheme by which the object is tested for membership in a closed substack.

Lemma 0.1. *Let $f : \mathcal{X} \to \mathcal{Y}$ be a morphism of algebraic stacks. Let $g : \mathcal{W} \to \mathcal{X}$ be a morphism of algebraic stacks which is surjective, flat, and locally of finite presentation. Then the scheme theoretic image of f exists if and only if the scheme theoretic image of $f \circ g$ exists and if so then these scheme theoretic images are the same.*

Proof. Assume $\mathcal{Z} \subset \mathcal{Y}$ is a closed substack and $f \circ g$ factors through $\mathcal{Z}$. To prove the lemma it suffices to show that f factors through $\mathcal{Z}$. Consider a scheme T and a morphism $T \to \mathcal{X}$ given by an object x of the fibre category of $\mathcal{X}$ over T. We will show that $f(x)$ is in fact in the fibre category of $\mathcal{Z}$ over T. Namely, the projection $T \times_{\mathcal{X}} \mathcal{W} \to T$ is a surjective, flat, locally finitely presented morphism. Hence there is an fppf covering $\{T_i \to T\}$ such that $T_i \to T$ factors through $T \times_{\mathcal{X}} \mathcal{W} \to T$ for all i. Then $T_i \to \mathcal{X}$ factors through $\mathcal{W}$ and hence $T_i \to \mathcal{Y}$ factors through $\mathcal{Z}$. Thus $f(x)|_{T_i}$ is an object of $\mathcal{Z}$. Since $\mathcal{Z}$ is a strictly full substack, we conclude that $f(x)$ is an object of $\mathcal{Z}$ as desired. □

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